

Advanced (Carolina Reaper)

Q1. If $f(x) = x + 2$ and $g(x) = x - 1$, what is $(f + g)(x)$?

- (A) $2x + 1$
- (B) $2x$
- (C) $x + 1$
- (D) x
- (E) $x - 1$

Q2. If $f(x) = x^2$ and $g(x) = 2x$, what is $(f - g)(x)$?

- (A) $x^2 - 2x$
- (B) $x^2 + 2x$
- (C) x^2
- (D) $2x$
- (E) x

Q3. If $f(x) = x - 3$ and $g(x) = x + 2$, what is $(f \cdot g)(x)$?

- (A) $x^2 - x - 6$
- (B) $x^2 + x + 6$
- (C) $x^2 - x - 2$
- (D) $x^2 + x - 2$
- (E) x^2

Q4. If $f(x) = x + 1$ and $g(x) = x - 2$, what is $(f \div g)(x)$?

- (A) $\frac{x+1}{x-2}$
- (B) $\frac{x-2}{x+1}$
- (C) $x + 3$
- (D) $x - 3$
- (E) x

Q5. If $f(x) = x^2 - 1$ and $g(x) = x + 1$, what is the domain of $(f + g)(x)$?

- (A) $(-\infty, -1) \cup (-1, \infty)$
- (B) $(-\infty, -1)$
- (C) $(-1, \infty)$
- (D) $(-\infty, \infty)$
- (E) $[-1, 1]$

Q6. If $f(x) = x - 2$ and $g(x) = x^2$, what is $(f \cdot g)(x)$?

- (A) $x^3 - 2x^2$
- (B) $x^3 + 2x^2$
- (C) $x^2 - 2x$
- (D) $x^2 + 2x$
- (E) x

Q7. If $f(x) = x + 3$ and $g(x) = x - 1$, what is $(f - g)(x)$?

- (A) $2x + 2$
- (B) $2x - 2$
- (C) $2x$
- (D) 4
- (E) 2

Q8. If $f(x) = x^2 + 1$ and $g(x) = x - 2$, what is $(f \div g)(x)$?

- (A) $\frac{x^2+1}{x-2}$
- (B) $\frac{x-2}{x^2+1}$
- (C) $x + 2$
- (D) $x - 2$
- (E) x

Open Ended Questions (FRQ)

Q9. If $f(x) = x^2 - 2x$ and $g(x) = x + 1$, find $(f + g)(x)$ and determine its domain.

Q10. If $f(x) = x - 1$ and $g(x) = x^2 + 1$, find $(f \cdot g)(x)$ and simplify the resulting expression.

Chef's Tips

- Remind students to apply the operations with functions carefully.
- Emphasize the importance of determining the domain of the combined functions.
- Encourage students to simplify the resulting expressions after operations.